1 (a) The reason why ReLU is not used since the b is strictly positive. If $b=0$, The log-likelihood will get problem. ($\log 0$ ×)



$\max\{0, x\}$

(b) loss: negative log likelihood.

$$L(\theta) = \prod_{i=1}^{n} p(y_i | x_i; \theta)$$

average negative log likelihood

$$L_{NLL}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} \log p(y_i | x_i, \theta)$$

Gaussian probability density function:

$$p(y_i | x_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(y_i - u_i)^2}{2\sigma_i^2}\right)$$

where $u_i = f_u(x_i)$

$\sigma_i = f_\sigma(x_i)$

$$p(y_i | x_i) = \frac{1}{\sqrt{2\pi} f_\sigma} \exp\left(-\frac{(y_i - f_u)^2}{2 f_\sigma^2}\right)$$

take negative log:

$$-\log\left[\frac{1}{\sqrt{2\pi} f_\sigma} \exp\left(-\frac{(y_i - f_u)^2}{2 f_\sigma^2}\right)\right]$$

$$= -\log\left[(2\pi)^{-\frac{1}{2}} f_\sigma^{-1} \exp\left(-\frac{(y_i - f_u)^2}{2 f_\sigma^2}\right)\right]$$

$$= -\left[-\frac{1}{2}\log(2\pi) + (-1)\log(f_\sigma) + \left(-\frac{(y_i - f_u)^2}{2 f_\sigma^2}\right)\right]$$

$$= \frac{1}{2}\log(2\pi) + \log(f_\sigma) + \frac{(y_i - f_u)^2}{2 f_\sigma^2}$$

take average:

$$L_{NLL} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{2}\log(2\pi) + \log f_\sigma + \frac{(y_i - f_u)^2}{2 f_\sigma^2}\right]$$